Attention Sink Probing in Diffusion Transformers: A Unified Analysis Framework

Author One[1], **Author Two**[1], **and Author Three**[2]

[1]University Alpha, [2]University Beta

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> **Key Finding:** Sink tokens in diffusion transformers are not merely artifacts—they serve a functional role in stabilizing the generation process. The accent primary color is used for emphasis throughout.

Our contributions are threefold: (1) we provide the first systematic analysis of attention sinks in diffusion architectures, (2) we establish a causal link between RoPE frequency structure and sink formation, and (3) we demonstrate practical applications through controlled sink manipulation.

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2.1. Theme Accent Colors

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2.2. Extended Palette

PaletteDeep | PaletteIndigo | PaletteMid | PaletteLight | PaletteSilver

Heatmap gradient:

0	10	20	30	40	50	60	70	80

3. Experimental Results

3.1. Quantitative Comparison

Table 1: Comparison of sink manipulation strategies on PRISM-Bench. **Bold**: best, underline: second best.

Method	Alignment ↑	Aesthetic ↑	FID ↓
Baseline	0.312	0.445	28.4
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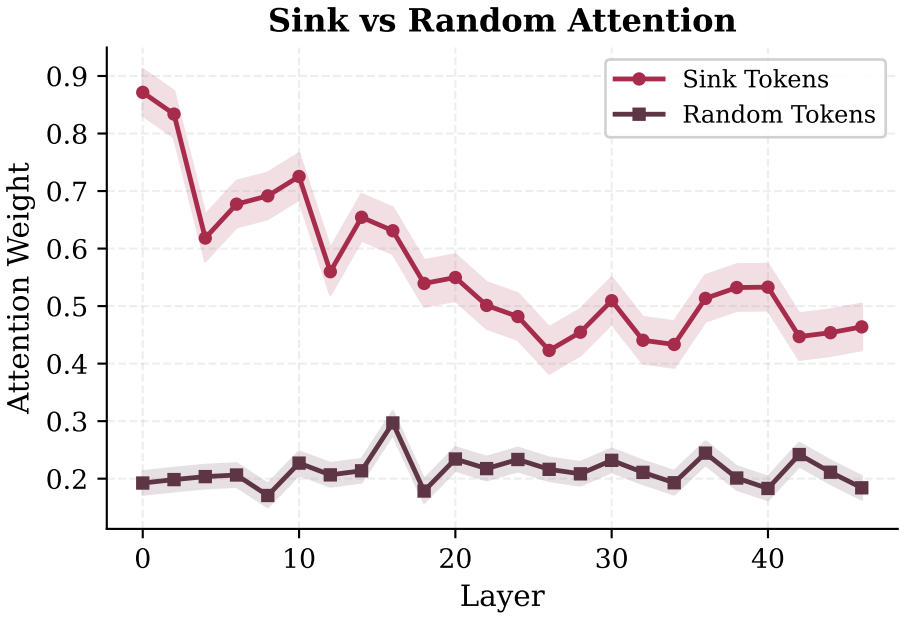
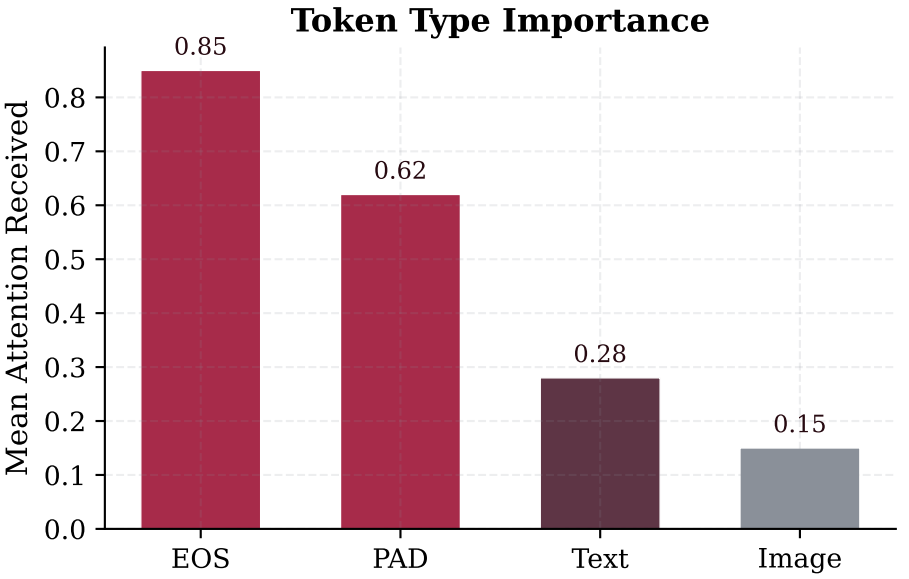
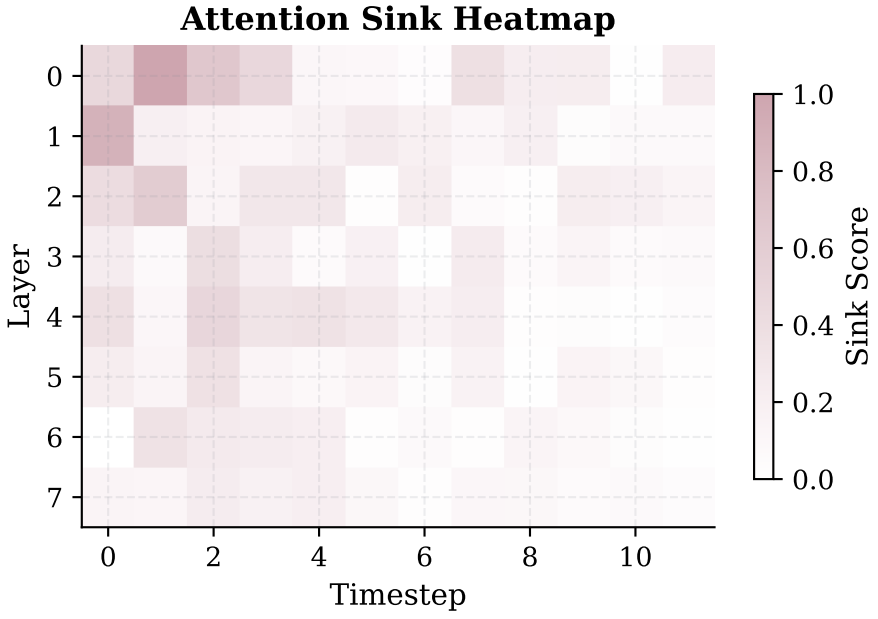
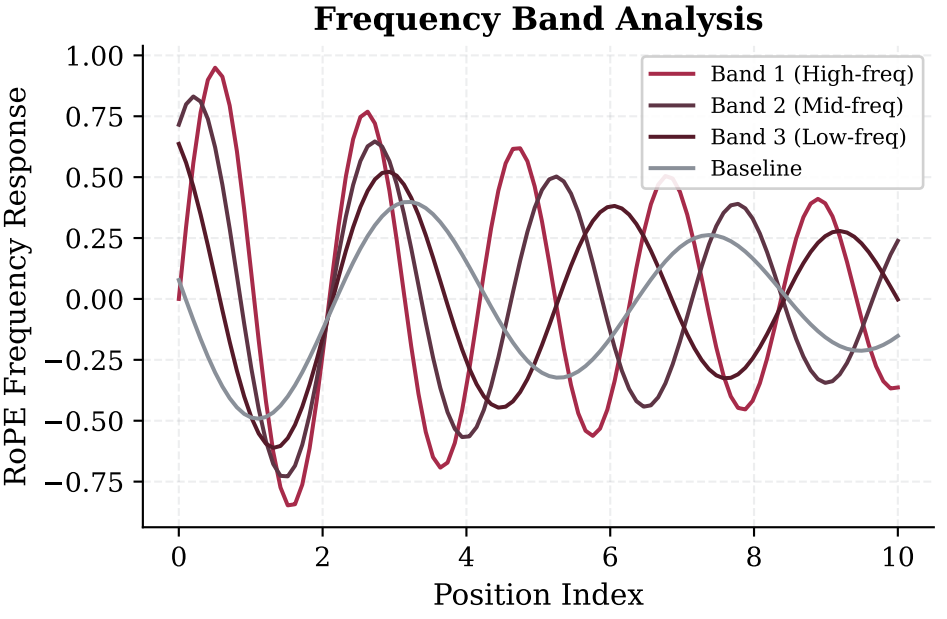
Red (Burgundy)









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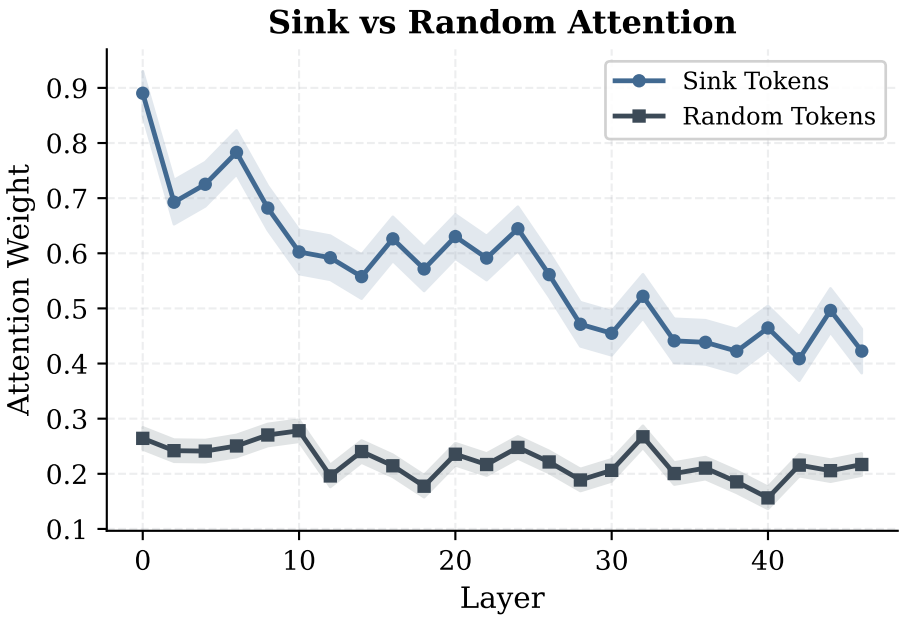

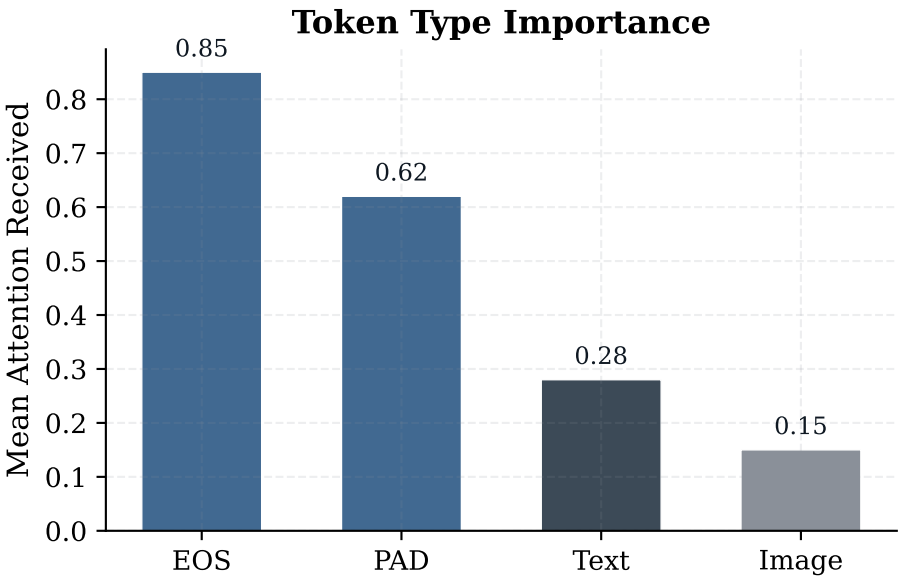

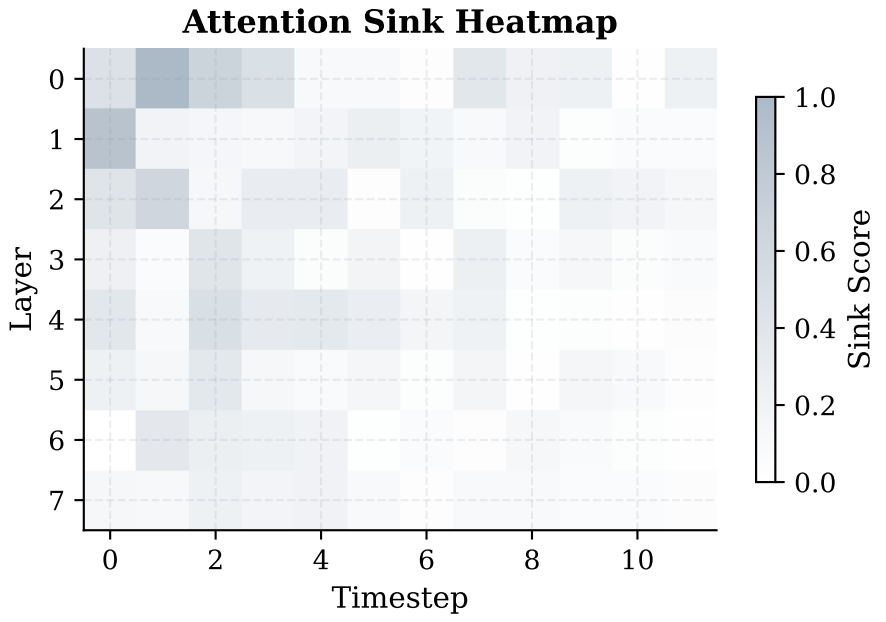

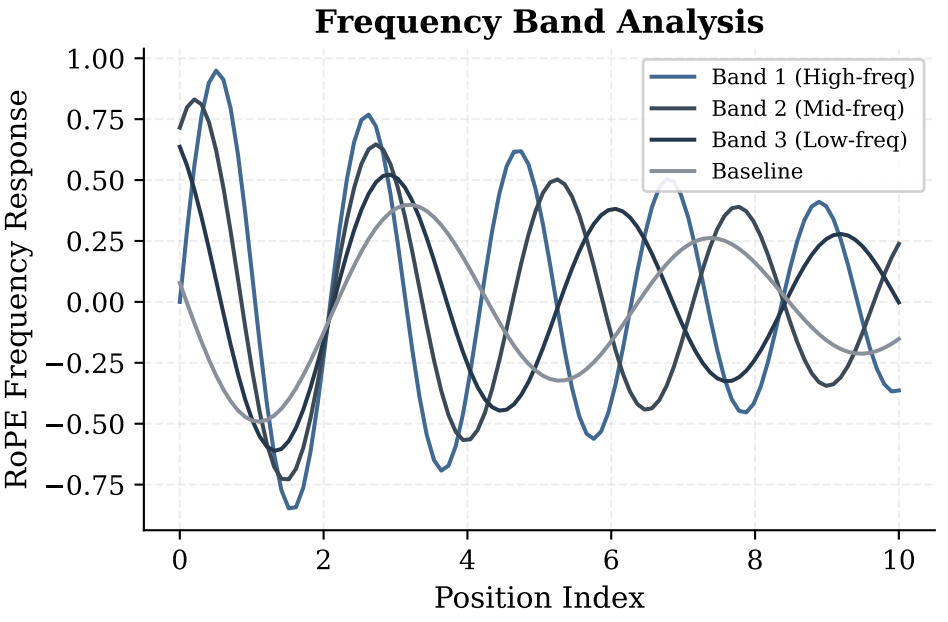

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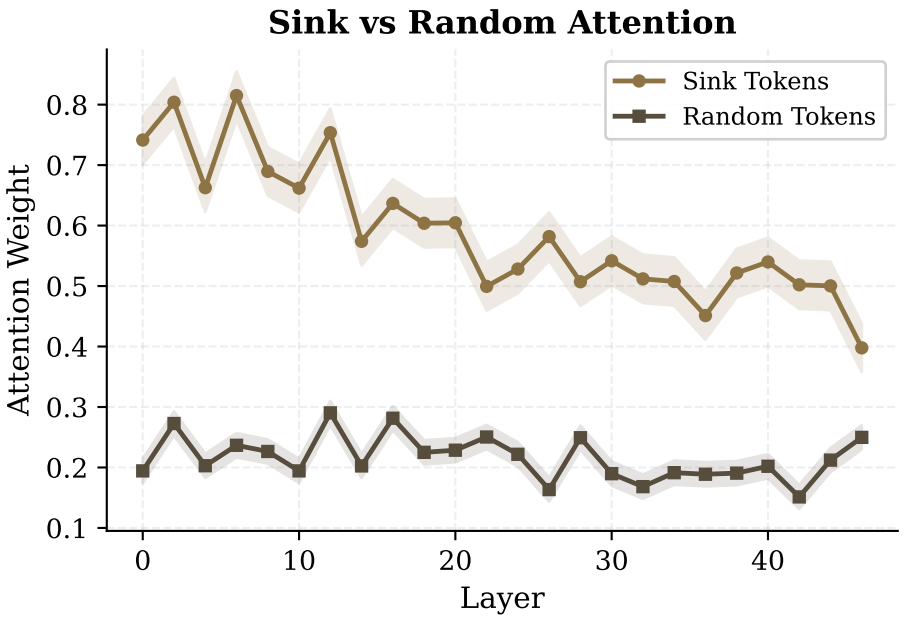
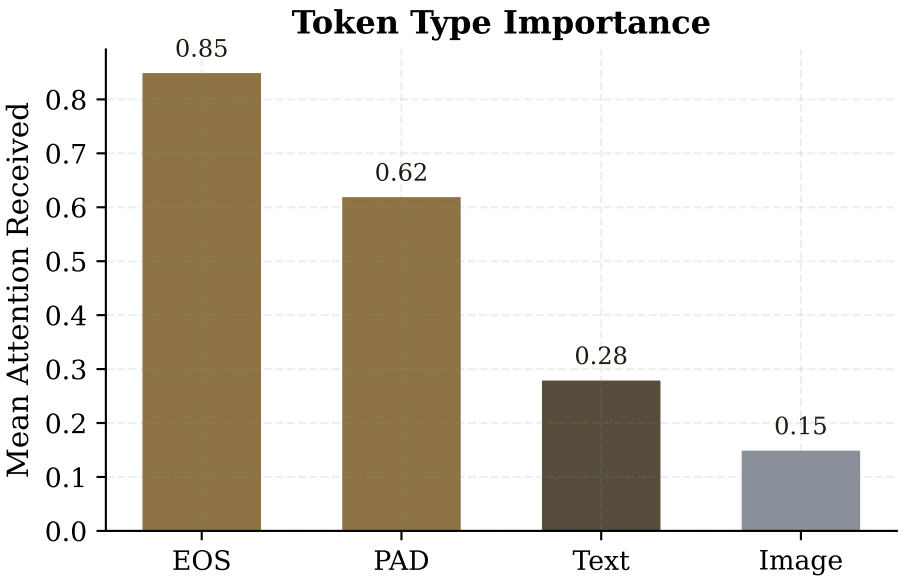
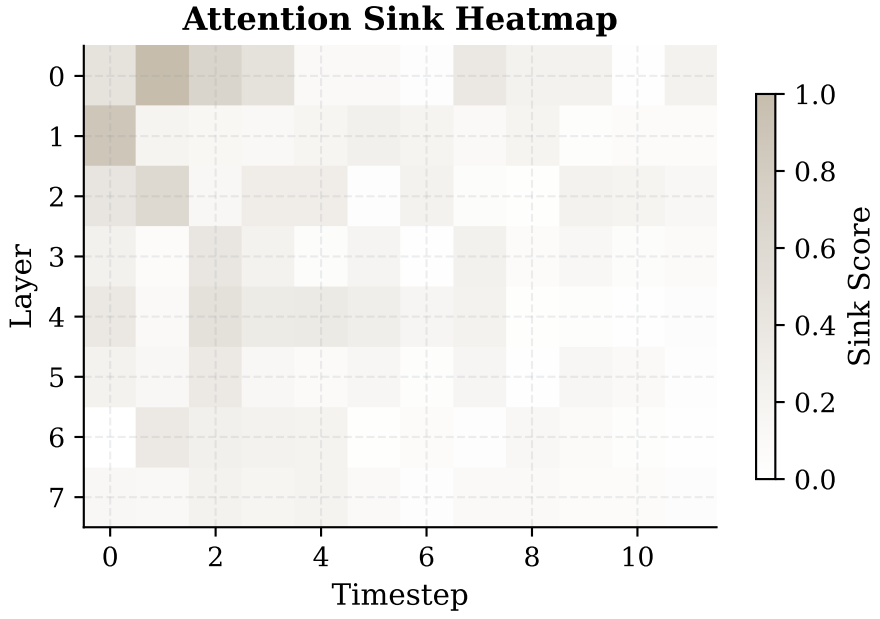
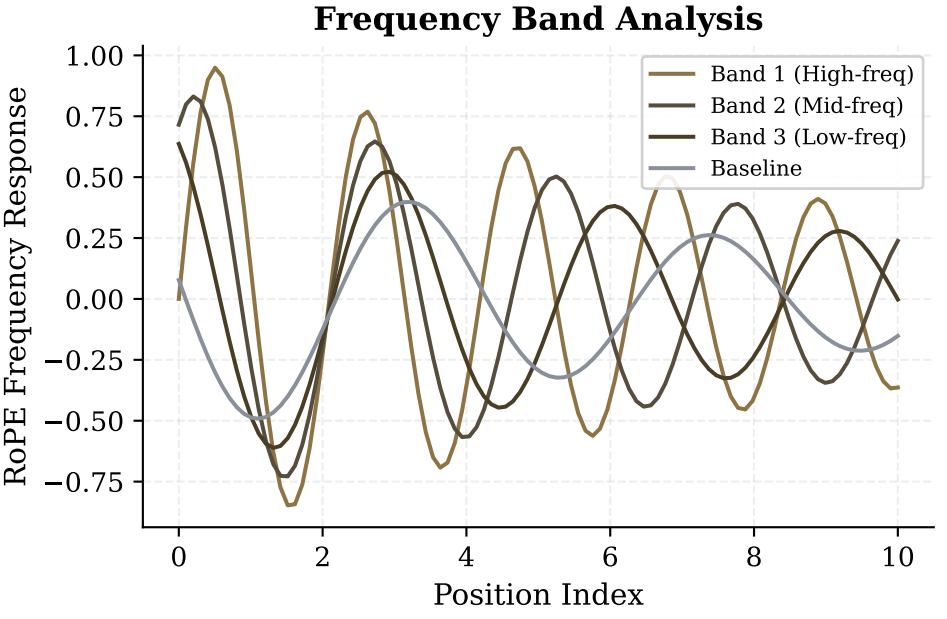
Gold (Warm Beige)









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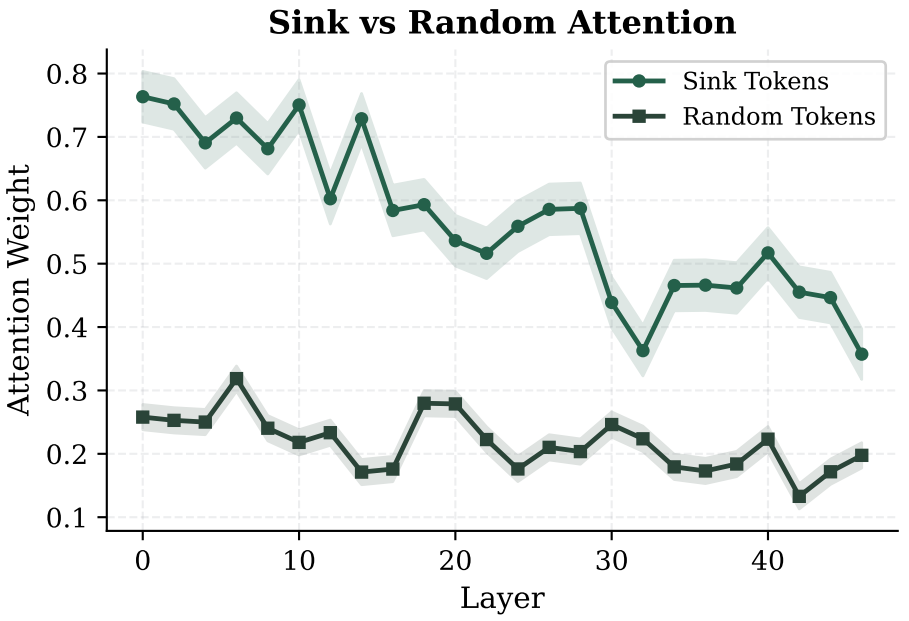
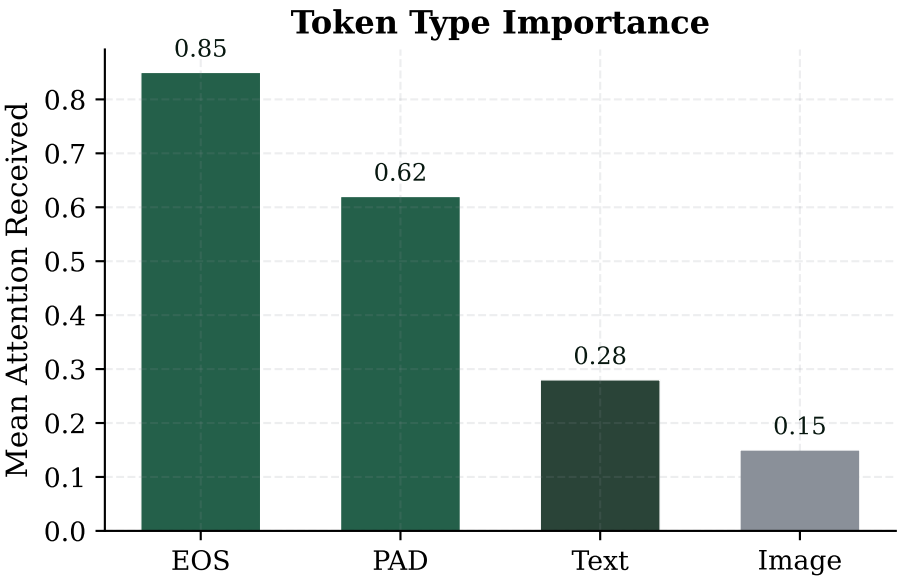
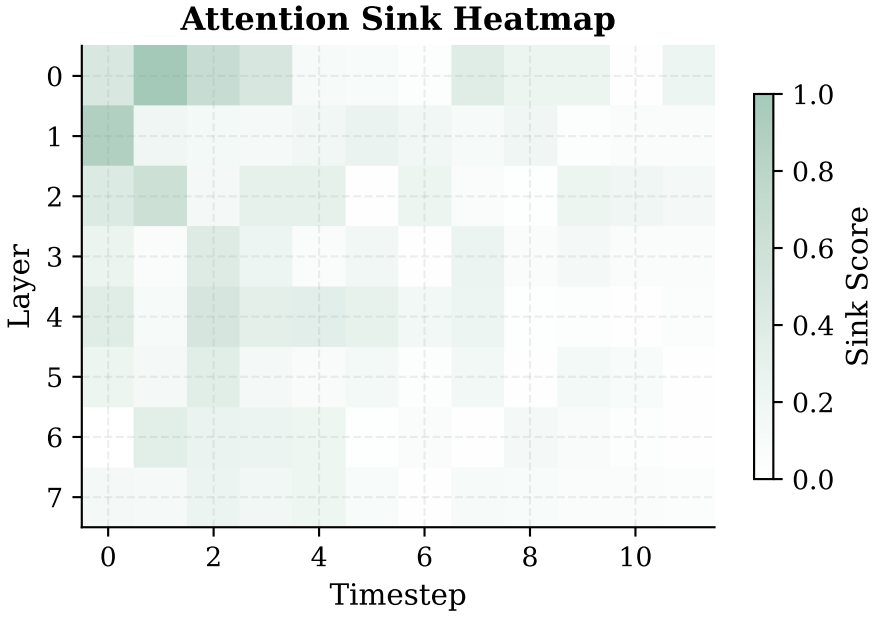
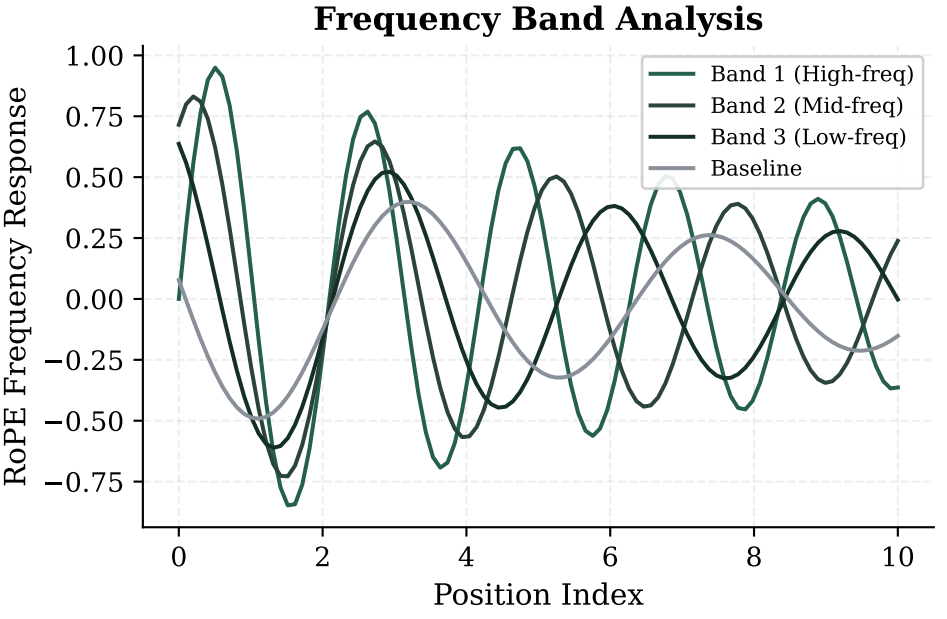
Green (Deep Forest)









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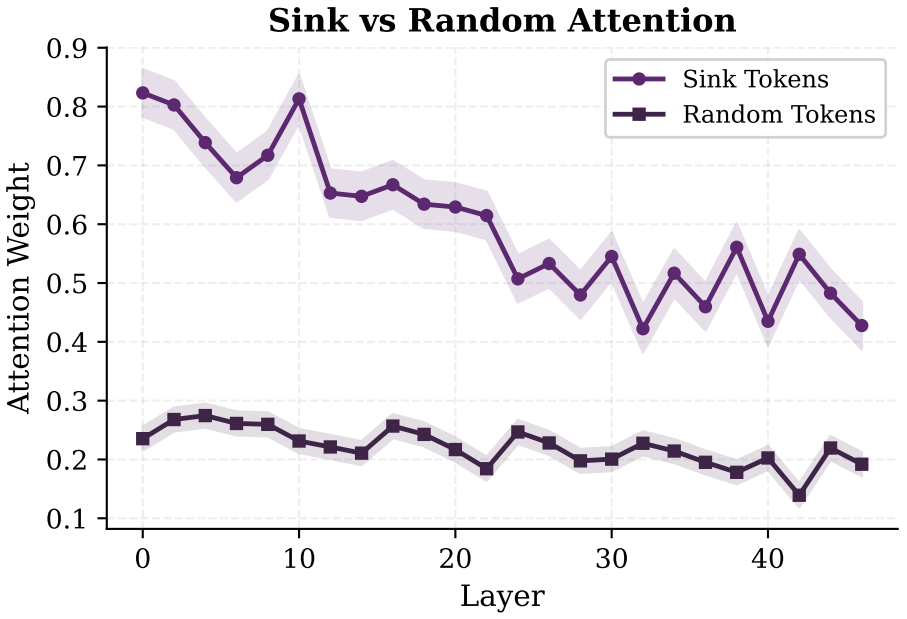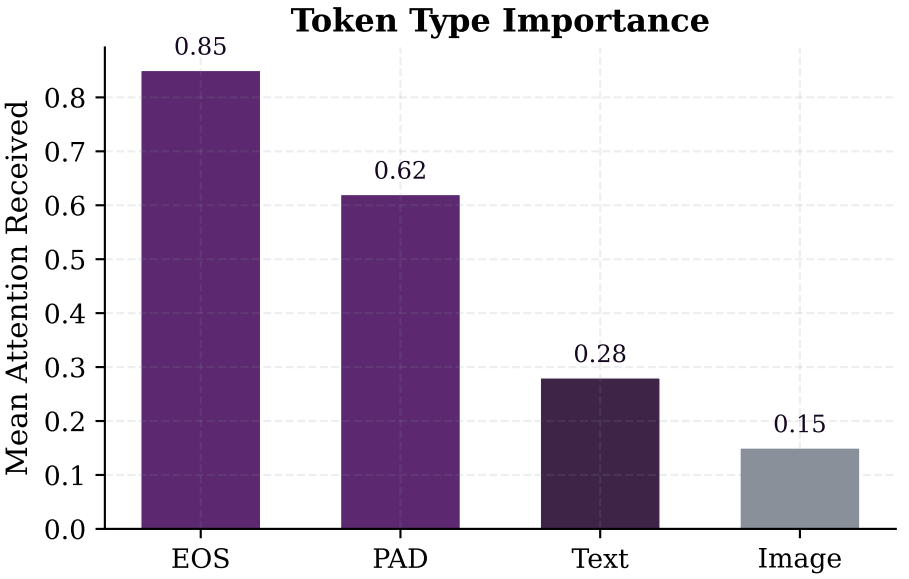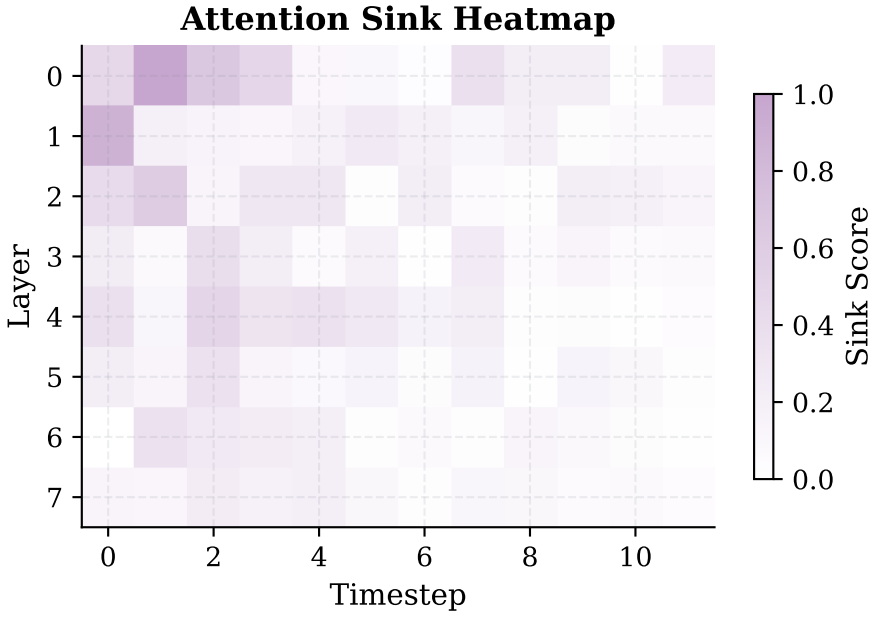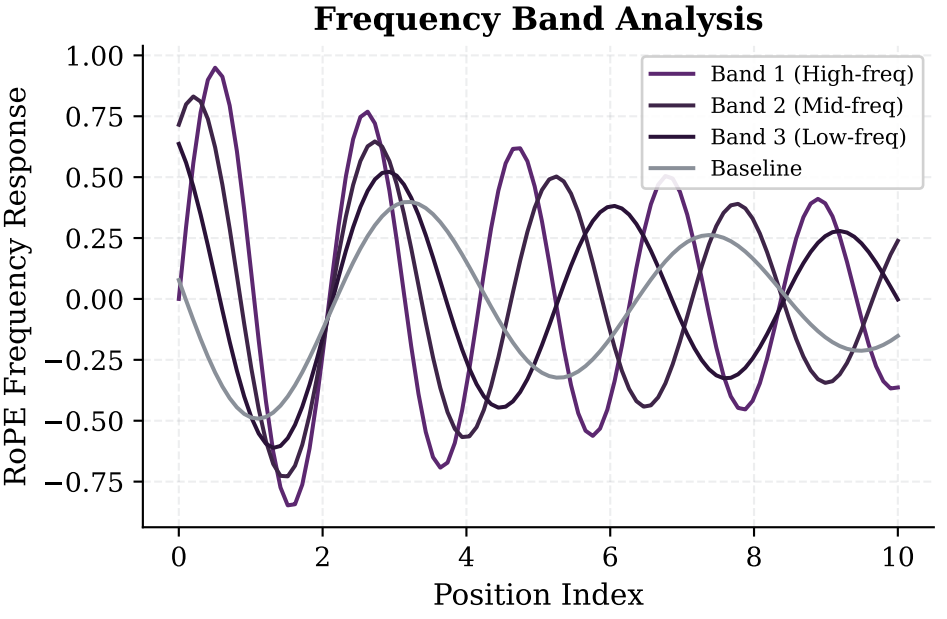
Purple (Deep Plum)